Supply Chain Analytics & Profitability Optimization

DataCo Supply Chain Case Study

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Tools: Python · Pandas · NumPy · Tableau · Excel · Prophet · PuLP · Business Intelligence

Executive KPI Summary

Key performance indicators highlighting the current state and overall operational health of the supply chain.

$36.78M	**$3.97M**
Total Revenue	Total Profit
11%	**65,752**
Profit Margin	Total Orders
20,652	**118**
Customers	Products

Executive Insights

- Revenue remained stable across 2015–2016 before declining significantly during Q4 2017.
- Overall profit margin remained approximately 11%, indicating opportunities to improve operational efficiency.
- A relatively small portfolio of 118 products generated nearly $36.8M in revenue, suggesting product concentration.
- More than 65,000 customer orders were fulfilled across multiple international markets.
- Customer analytics identified high-value customer segments suitable for retention strategies.
- Optimization models recommended prioritizing profitable products, improving warehouse utilization, and focusing on high-performing markets.

Business Impact

IMPROVED DECISION MAKING | INVENTORY OPTIMIZATION | DEMAND FORECASTING | SUPPLY CHAIN PROFITABILITY

Executive Summary

Objective

Develop an end-to-end analytics solution to improve profitability, inventory planning, logistics performance, and strategic decision-making using the DataCo Supply Chain Dataset.

Business Outcomes

- Revenue Analysis
- Customer Segmentation
- Product Analytics
- Logistics Performance
- Demand Forecasting
- Linear Programming Optimization
- Executive Dashboard
- Business Recommendations

Business Problem

The company faces several operational challenges.

- Declining profitability
- Increasing delivery delays
- High inventory costs
- Poor visibility into customer behaviour
- Limited demand forecasting
- Lack of optimization-based decision making

Business Goal: Transform raw transactional data into actionable business insights that improve profitability and supply chain performance.

Dataset Overview

Dataset

DataCo Supply Chain Dataset

Contains

- Orders
- Customers
- Products
- Shipping
- Markets
- Categories

Project Scope

- Data Cleaning
- Feature Engineering
- KPI Analysis
- Exploratory Data Analysis
- Customer Analytics
- Product Analytics
- Logistics Analytics
- Forecasting
- Optimization

Project Roadmap

Data Preparation

Data Cleaning

- Missing value treatment
- Duplicate handling
- Datatype correction
- Data validation

Feature Engineering

- Unit Profit
- Profit Margin
- Order Value Segment
- Average Selling Price
- Delivery Delay
- Shipping Performance

Exploratory Data Analysis

Analysed

Revenue Trends

Regional Performance

Category Performance

Monthly Sales

Seasonality

Key Findings: Highest revenue markets contribute the majority of business sales while some categories generate significantly lower profits.

Product Analytics

Analysed

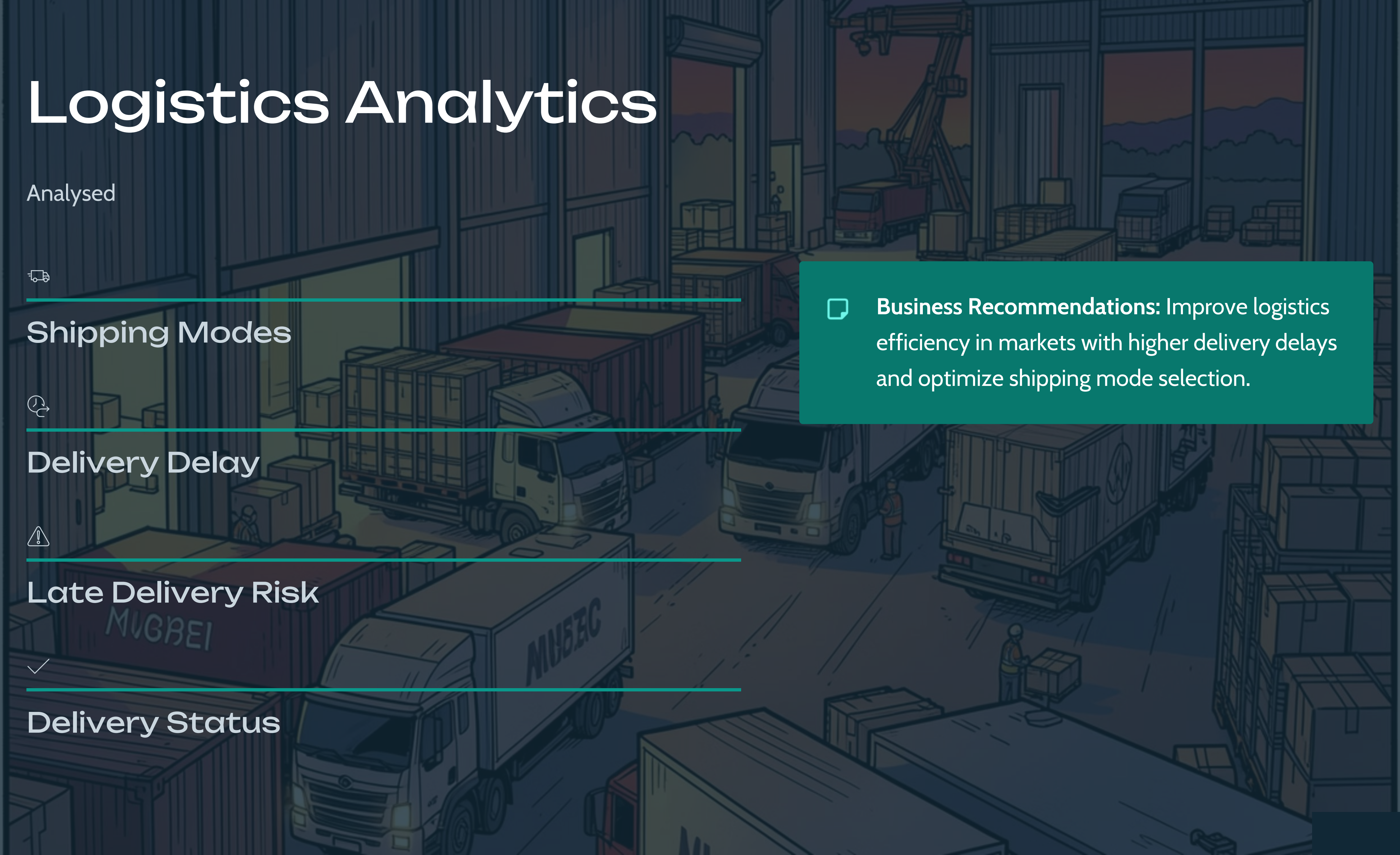
Top Products

Bottom Products

Discount versus Profit

Product Profitability

Business Recommendations: Increase inventory allocation for high-profit products while reviewing pricing strategies for low-performing products.



Logistics Analytics

Analysed

Shipping Modes

Delivery Delay

Late Delivery Risk

Delivery Status

Business Recommendations: Improve logistics efficiency in markets with higher delivery delays and optimize shipping mode selection.

Customer Analytics

Business Value: Identify loyal customers, at-risk customers, and opportunities to improve customer retention.

Performed

RFM Analysis	Customer Segmentation
Cohort Analysis	

ABC Analysis

Explained

Business Recommendation: Prioritise inventory investment in Class A products.

1

Class A

High Revenue Products

2

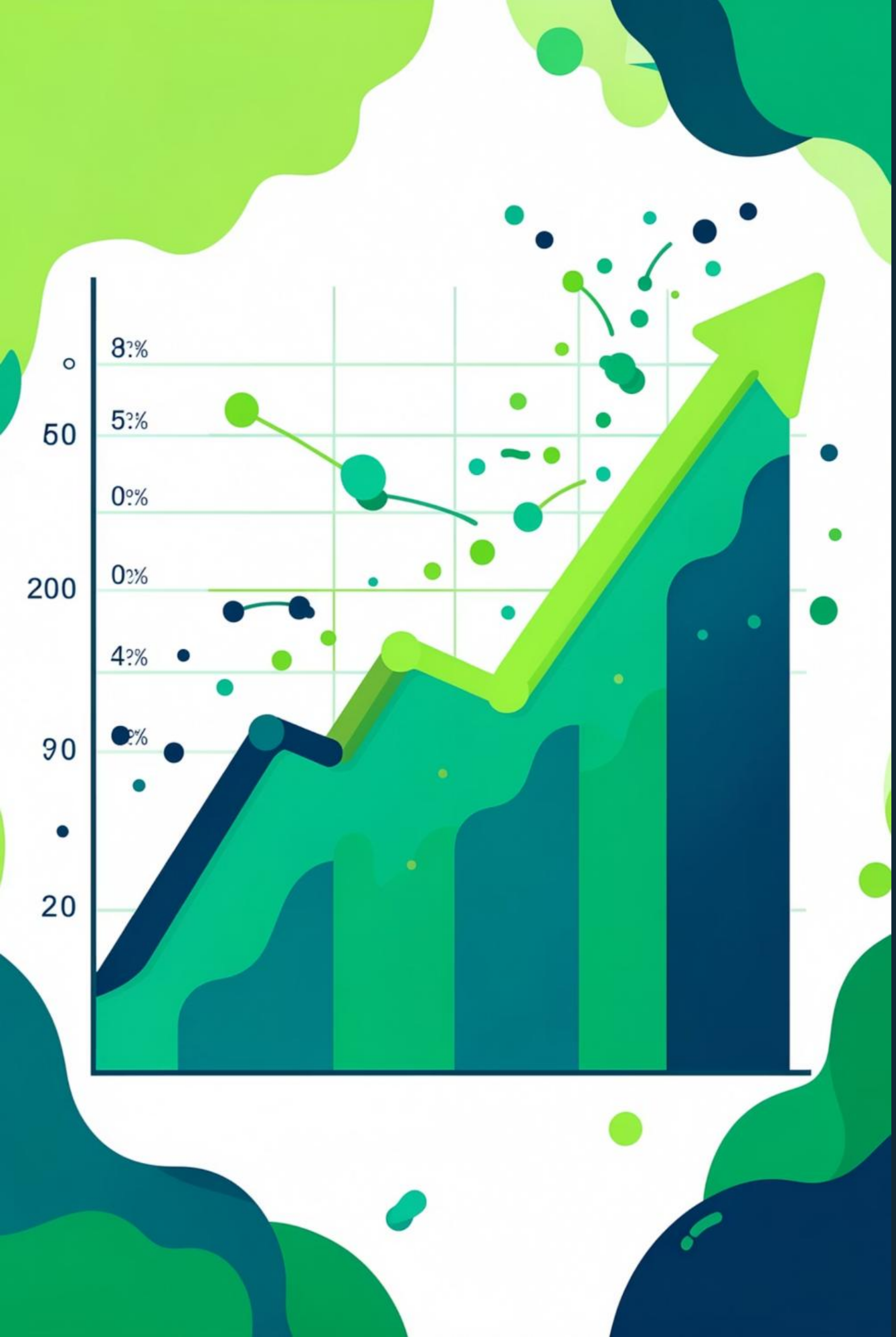
Class B

Medium Revenue Products

3

Class C

Low Revenue Products



Demand Forecasting

Accurate prediction of future demand is critical for optimising supply chain operations and ensuring product availability.

Forecasting Model

Leveraging **Facebook Prophet** for robust and accurate time series forecasting.

Forecast Output

Predicting future monthly demand to proactively anticipate market needs and optimise stock levels.

Optimisation Impact

Crucial for improving procurement planning, inventory allocation, and efficient resource management.

Business Value: Enhance efficiency and reduce costs through improved procurement planning, inventory allocation, and resource planning.

Supply Chain Optimization

Applying advanced mathematical techniques to maximize efficiency and profitability across the supply chain.

Technique

Linear Programming, implemented using the **PuLP** Python library for robust, scalable optimization.

Objective

To strategically prioritise products and markets, considering profitability, margins, demand, and operational constraints.

Decision Variables

Optimizing allocation and flow for individual products and across various market segments.

Constraints

Incorporating limitations such as warehouse capacity, inventory levels, desired profitability, and required delivery performance.

Business Value: Drive strategic decision-making by optimizing resource allocation to achieve maximum profit margins and operational efficiency.

Detailed Scenario Analysis

Evaluating different supply chain strategies to identify the most effective approach for improving profitability, inventory utilisation, and market performance under real-world business constraints.

Scenario 1: Product Optimization

Business Problem

Not every product contributes equally to profitability. Investing in low-performing products reduces overall business efficiency and ties up valuable capital.

Optimization Goal

To select the optimal mix of products that maximises overall business value and ensures sustainable growth.

Objective Function

Maximise total business score based on:

- Revenue Generation
- Profit Realisation
- Achievable Profit Margin

Business Value

- Prioritise high-profit, high-demand products
- Reduce investment in low-performing or unprofitable products
- Improve overall product portfolio profitability and market share

Scenario 2 – Inventory Capacity Constraint

Business Problem

Warehouse capacity is limited, making it impossible to stock every product.

Optimization Goal

Select products while ensuring total inventory remains within warehouse capacity.

Constraints

- Inventory capacity limit
- Product quantity constraint
- Profitability optimization

Business Value

- Better warehouse utilization
- Lower inventory holding costs
- Efficient allocation of storage space
- Improved inventory planning

Scenario 3 – Market Optimization

This scenario focuses on strategically prioritizing markets to maximize overall business performance by considering various financial and operational metrics.

Business Problem

Not all markets offer the same revenue and profit potential. Some markets also incur higher delivery delays and operational costs, reducing overall efficiency.

Optimization Goal

To select and prioritize markets that maximise business performance while maintaining operational feasibility.

Decision Factors

- Revenue
- Profit
- Profit Margin
- Delivery Performance
- Late Delivery Risk

Business Value

- Focus resources on profitable markets
- Improve logistics efficiency
- Reduce operational costs
- Support strategic market expansion

Optimization Scenario Comparison

Comparison of optimization strategies and their impact on key business metrics.

Metric	Current Business	Product Optimization	Inventory Optimization	Market Optimization
Revenue	$36.78M	$29.84M	$28.24M	$31.46M
Profit	$3.97M	$3.21M	$3.04M	$3.39M
Orders	65,752	58,893	56,838	56,746
Demand (Units)	384,079	230,433	192,034	324,251

Key Insights

- Product Optimization selected the highest-value products, reducing overall demand while maintaining strong profitability.
- Inventory Optimization achieved the largest reduction in inventory demand, making it suitable for warehouse capacity constraints.
- Market Optimization retained the highest revenue and profit among the optimization scenarios while reducing operational scale.
- All three optimization models reduced the number of orders and demand, reflecting a strategic focus on higher-value business opportunities rather than maximizing transaction volume.

Business Recommendation

- Use Product Optimization for portfolio rationalization and product prioritization.
- Apply Inventory Optimization when warehouse space or storage costs are the primary constraint.
- Use Market Optimization for strategic regional expansion and resource allocation because it preserves the highest revenue and profit among the optimized scenarios.

Scenario Analysis

Strategic scenario planning enables proactive evaluation of different business decisions before implementation, minimizing risk and maximizing potential outcomes across the supply chain.

Scenario 1: Current Business

Reflects the baseline operational model, highlighting existing practices and performance levels.

Scenario 2: Inventory Constraint

Optimization efforts focused on reducing inventory carrying costs and improving cash flow efficiency.

Scenario 3: Market Optimization

Explored strategies to maximise profitability by strategically allocating resources across products and markets.

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Scenario 4: Product Optimization

Focuses on strategic product selection and portfolio optimization to maximize revenue and profitability across the supply chain.

Business Recommendation: Scenario 3, focusing on market optimization, yields the highest profitability, suggesting a strategic shift towards product and market resource allocation.

Interactive Tableau Dashboard

A dynamic, user-friendly Tableau dashboard provides real-time insights across various operational areas, empowering executives with actionable data.

Key Functionality: Intuitive filters and drill-down capabilities within each dashboard allow executives to explore data granularly, fostering agile and data-driven decision-making.

Executive Dashboard

High-level overview of critical business KPIs for strategic decision-making.

Sales Dashboard

Detailed insights into sales performance, revenue trends, and market share.

Product Dashboard

Performance metrics for individual products, categories, and profitability.

Logistics Dashboard

Tracking shipping efficiency, delivery times, and associated costs.

Customer Dashboard

Insights into customer behaviour, segmentation, and retention strategies.

Forecast Dashboard

Visualizing demand predictions, accuracy, and future market needs.

Optimization Dashboard

Displaying results and recommendations from supply chain optimization models.

RECOMMENDATIONS

Business Recommendations

Based on our analysis, we propose the following strategic recommendations to enhance supply chain efficiency and profitability:

Optimize Inventory

Increase allocation for high-profit products and prioritise Class A inventory to meet demand effectively.

Market Expansion

Expand investment and focus on profitable markets identified through detailed analytics.

Proactive Planning

Utilise demand forecasts for more accurate procurement planning and resource allocation.

Enhance Logistics

Improve delivery performance in markets experiencing high delays to boost customer satisfaction.

Strategic Pricing

Review and reduce discounts on low-margin products to improve overall profitability.

Customer Focus

Implement targeted campaigns to retain high-value customers, nurturing loyalty and repeat business.

Leverage interactive Tableau dashboards for real-time KPI monitoring and agile decision-making.

Key Takeaway: Implementing these data-driven strategies will lead to improved operational efficiency and a significant uplift in profitability.

Business Impact

Driving Value Across the Supply Chain

Improved Profitability

Maximising revenue and reducing costs for higher margins.

Better Inventory Utilisation

Optimising stock levels to meet demand efficiently and reduce carrying costs.

Improved Logistics Planning

Streamlining transportation and distribution for faster, more reliable deliveries.

Enhanced Customer Retention

Boosting satisfaction through consistent product availability and timely service.

Clear Demand Visibility

Gaining foresight into future market needs to proactively adjust operations.

Data-Driven Decisions

Empowering strategic choices with actionable insights from comprehensive analytics.

Operational Efficiency

Optimising workflows and resource allocation for peak performance.

Strategic Planning

Aligning supply chain operations with overall business goals for sustainable growth.

Future Scope

Our proposed future initiatives aim to further enhance supply chain capabilities and leverage cutting-edge technology for sustained growth and efficiency.

Real-time Dashboard Integration

Implement dynamic dashboards offering immediate insights into supply chain performance.

Cloud Deployment

Migrate analytical tools and data infrastructure to a scalable, secure cloud environment.

Machine Learning Demand Prediction

Integrate advanced ML models for more accurate and adaptive demand forecasting.

Route Optimisation

Develop algorithms to dynamically optimise delivery routes, reducing costs and transit times.

Inventory Optimisation

Utilise AI-driven strategies to minimise stockouts and overstocking, improving working capital.

Supplier Optimisation

Enhance supplier relationships and performance through data-driven evaluation and collaboration.

ERP Integration

Seamlessly integrate analytical insights with existing Enterprise Resource Planning systems.

AI-powered Recommendation Engine

Implement intelligent systems to provide proactive recommendations for strategic decisions.

Conclusion

This project demonstrates a complete end-to-end supply chain analytics solution, integrating several key areas to support strategic business decisions:

Business Intelligence	Customer Analytics
Demand Forecasting	Optimisation

This comprehensive approach is designed to drive significant improvements in supply chain efficiency and profitability.

Thank You